Probabilistic Modelling Coursework 1

Deadline: 14th March 2025 6PM

Instructions:

1) You need to submit a **single** PDF file
2) You are free to use any programming language to answer questions, but you **MUST** show evidence of code that was used. This can be screenshots of the code (see "Sample_Solution.pdf")
3) You will **NOT** be marked for your coding ability, but you **MUST** show evidence of code that was used.
4) Your coursework will **NOT** be marked if you do not include the proof of code used to answer the questions
5) The mark breakdown for the assessment is:

    Question 1: 25 marks

    Question 2: 25 marks

    Question 3: 25 marks

    Presentation of report: 15 marks

    Total: 90 marks

6) Your presentation will be marked down if:
    - Figures do not have the appropriate axis
    - Figures are not labelled
    - Equations are not labelled appropriately
7) Note that there is no word count limit/page limit. So, show all your working out.

By submitting my assignment:

I declare that I have personally prepared this assignment. The work is my own, carried out personally by me unless otherwise stated and has not been generated using Artificial Intelligence tools unless specified as a clearly stated approved component of the assessment brief. All sources of information, including quotations, are acknowledged by means of the appropriate citations and references. I declare that this work has not gained credit previously for another module at this or another University. I understand that plagiarism, collusion, copying another student and commissioning (which for the avoidance of doubt includes the use of essay mills and other paid for assessment writing services, as well as unattributed use of work generated by Artificial Intelligence tools) are regarded as offences against the University's Assessment Regulations and may result in formal disciplinary proceedings.

I understand that by submitting this assessment, I declare myself fit to be able to undertake the assessment and accept the outcome of the assessment as valid.

Question 1 [25 marks]

You have been tasked with modelling the number of babies born every hour in different hospitals worldwide. 10,000 data points have been collected and stored in "Hospital.csv" The likelihood of each data point is:

$$P(x_i|\lambda) = \frac{e^{-\lambda}\lambda^{x_i}}{x_i!}, \text{for } i = 1, \dots, 10000$$

Here x_i represents each data point, and λ is a random variable with the following prior distribution:

$$P(\lambda|\alpha, \beta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}, \lambda > 0$$

where $\Gamma(\alpha)$ is a gamma function, α and β are some parameters.

a) Import the data "Hospital.csv" and plot the histogram of the data. Comment on what you see.
[3 marks]

b) Plot $P(\lambda|\alpha, \beta)$ over $0 < \lambda \leq 10$ with $\alpha = 2$ and $\beta = 1$
[4 marks]

c) By using the data, compute the maximum likelihood value (λ_{ML}) for λ
[3 marks]

d) Plot $P(x|\lambda_{ML})$ over $x = \{0,1,2,3,4,5,6,7,8,9\}$
[4 marks]

e) By using the prior and likelihood, compute the posterior distribution of $P(\lambda|x_1, \dots, x_{10000}, \alpha, \beta)$
[4 marks]

f) By using the posterior distribution, compute the Maximum a-posterior value (λ_{MAP}) for λ with $\alpha = 2$ and $\beta = 1$
[4 marks]

g) Plot the posterior distribution found in e) over an appropriate domain with $\alpha = 2$ and $\beta = 1$
[3 marks]

Question 2 [25 marks]

Some data has been collected on the number of hours a student spends on homework per week. It is believed that the students can be classified into two classes:

- Class 1: A student failed the module
- Class 2: A student passed the module

You have been given some data on 2005 students ("Students.csv") which consists of two columns

- x: number of hours spent on homework
- class: if the student belongs to Class 1, Class 2 or Unknown

Note that the Unknown class will be used in part e). A second binary random viable $z = \{1,2\}$ is used to classify the students, if $z_i = 1$ then the ith student failed the module, otherwise they passed the module.

It is assumed that:

$$P(x|z = 1) = \text{uniform}(a, b)$$

where a and b are constants.

$$P(x|z = 2) = \text{Normal}(\mu, \sigma^2)$$

where μ and σ are constants and it is also assumed that the prior of passing or failing a module is equal.

a) Import the data "Students.csv" and plot the data
   [2 marks]
b) Using any appropriate method compute the value of a and b
   [2 marks]
c) Using any appropriate method compute the value of μ and σ
   [3 marks]
d) Plot the densities of $P(x|z = 1)$ and $P(x|z = 2)$ on the same plot
   [4 marks]
e) Classify the students who have "Unknown" in the class column
   [4 marks]
f) Find the decision boundary x_c, such that $P(x_c|z = 1) = P(x_c|z = 2)$ you may round the parameters you found in part b) and c) to integers for simplicity
   [5 marks]

g) Compute the misclassification error of classifying a student who failed as a student who passed
[5 marks]

Question 3 [25 marks]

James is concerned that he has contracted a new disease ($J = 1$). He shows suspicious symptoms: a sore throat ($S = 1$) and an elevated fever ($F = 1$). He has been in touch only with Ruth (R) over the last week and thinks she has infected him. The following probabilities are given:

- The prior probability of Ruth having the disease is $P(R = 1) = \frac{1}{4}$
- Ruth spent a lot of time with James recently and is likely to have infected him if she is infectious $P(J = 1|R = 1) = \frac{3}{4}$; clearly, she has not infected him if she does not have the disease
- A person (say James) who has contracted the disease, always experiences a sore throat $P(S = 1|J = 1) = 1$ but has elevated fever only in half of the cases $P(F = 1|J = 1) = \frac{1}{2}$; not having contracted the disease, James can still have elevated fever $P(F = 1|J = 0) = \frac{1}{4}$ and a sore throat $P(S = 1|J = 0) = \frac{1}{4}$

You must keep all probabilities in fractions:

a) Plot the directed acyclic graphical model that represents the problem
[2 marks]

b) Write an expression for the joint probability of all variables in the problem as well as the probabilities given in the question
[3 marks]

c) What is the probability that Ruth has the disease given that James has a sore throat and elevated fever $P(R = 1|S = 1, F = 1)$?
[8 marks]

d) After a while, James drinks some water and realises that his sore throat was a result of not drinking enough so that S = 0 . What is the probability that Ruth has the disease now?
[7 marks]

e) Later in the day James is again unclear whether he has a sore throat or not. A lateral flow test (T) shows that Ruth is positive for the disease ($T = 1$). However, these tests are inaccurate and the probability of a person having the disease testing positive is $\frac{3}{4}$, but also healthy individuals may test positive with probability $\frac{1}{4}$. Given

that James has elevated fever ($F = 1$), what is the probability that he is ill?
[5 marks]